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Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: B.E., CSE, V Semester

Course Code & Title: CCS375 Web Technologies

Name of the Faculty member: Dr. S. Poornam

Innovative Practice Description

Unit / Topic: Unit II/ Event handling
Course Outcome: CO2
Topic Learning Outcome: TLO4
Activity Chosen: Think Pair Share
Date & Time: 28.08.2025 & 10:30 A.M

Justification

This innovative classroom activity helps students visualize how JavaScript handles user interactions such as mouse clicks, key presses, and form submissions. Instead of only seeing code on the screen, students physically act out the process of event generation, propagation, and handling, making an abstract programming concept easy to grasp. The activity promotes active learning, collaboration, and conceptual clarity, bridging the gap between JavaScript syntax and real-world event-driven web applications.

Time Allotted for the Activity: 20 Minutes

Details of the Implementation

- The instructor introduced the concept of JavaScript event handling, explaining the roles of the Event Source, Event Object, Event Listener, and the Browser environment.
- During the individual reflection phase, students considered what occurs when a user clicks a button on a webpage.

- Students thought about how the browser determines which function to execute in response to that click. Additionally, students reflected on the concepts of event bubbling and event capturing, predicting the sequence in which events flow through the document.

Think Phase:

- Each student individually reflected on the process that occurs when a user clicks a button on a webpage.
- They considered how the browser identifies and executes the appropriate function in response to the click.
- Students also explored the concepts of event bubbling and event capturing, analyzing how events propagate through the DOM hierarchy.
- They expressed their understanding through short written answers or diagrams predicting the sequence of the event flow.

Pair Phase:

- Students paired up to compare their answers and reasoning.
- Each pair discussed differences in their understanding and refined their explanation of how an event travels from the source to the listener.
- The instructor then selected a few pairs to demonstrate the process physically to reinforce concepts.
  - Student 1 = Button (Event Source)
  - Student 2 = Event Object (carries information like type and target)
  - Student 3 = Event Listener (Handler Function)
  - Student 4 = Browser (Environment managing events)

Share Phase:

- Pairs shared their final understanding with the class, supported by a visual demonstration:
- Placing text behind the image (hidden effect).
- Bringing the background forward (illogical in real design).

CO – PO / PSO Mapping

CO	PO1	PO2	PO3	PO4	PSO1
CO1	2	2	3	2	2

(1 – Low, 2 – Moderate, 3 – High)

Innovative practice	PO1	PO2	PO3	PO4	PSO1
	2	2	3	2	2
Justification for correlation	Students gained fundamental understanding of how JavaScript reacts to user interactions.	They analyzed event-driven behavior to identify when and where handlers should be attached.	Students implemented simple interactive webpages responding to user input using event listeners.	Students debugged event flow issues (e.g., bubbling vs capturing conflicts).	They applied knowledge to design interactive and responsive web interfaces.

PO/PSO mapped: PO1, PO2, PO3, PO4, PSO1

Images / Screenshot of the practice:

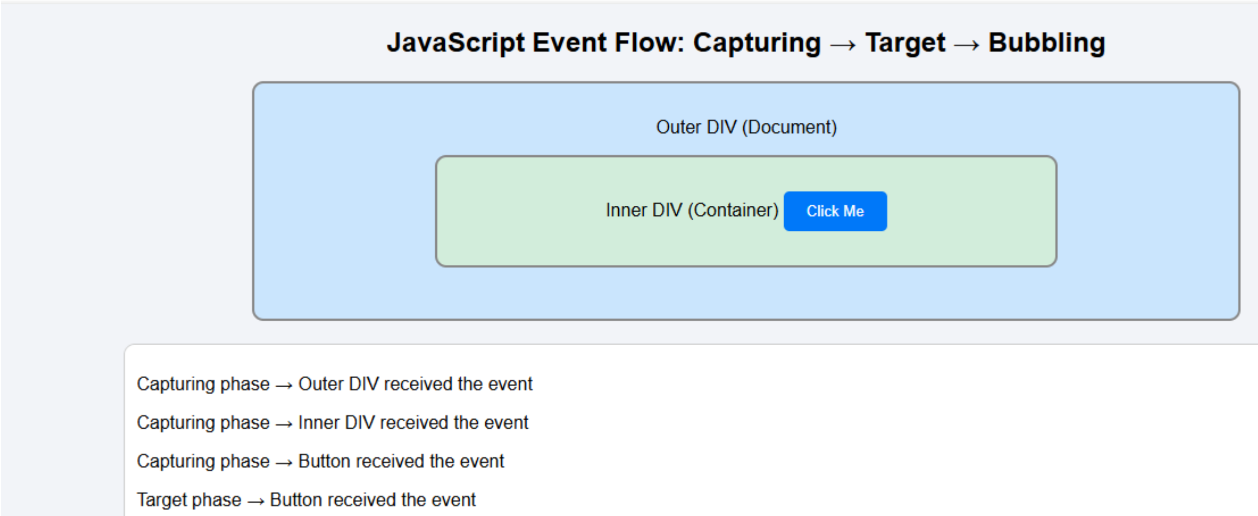

Fig:1. Think share pair activity by Sri Prishigaa R

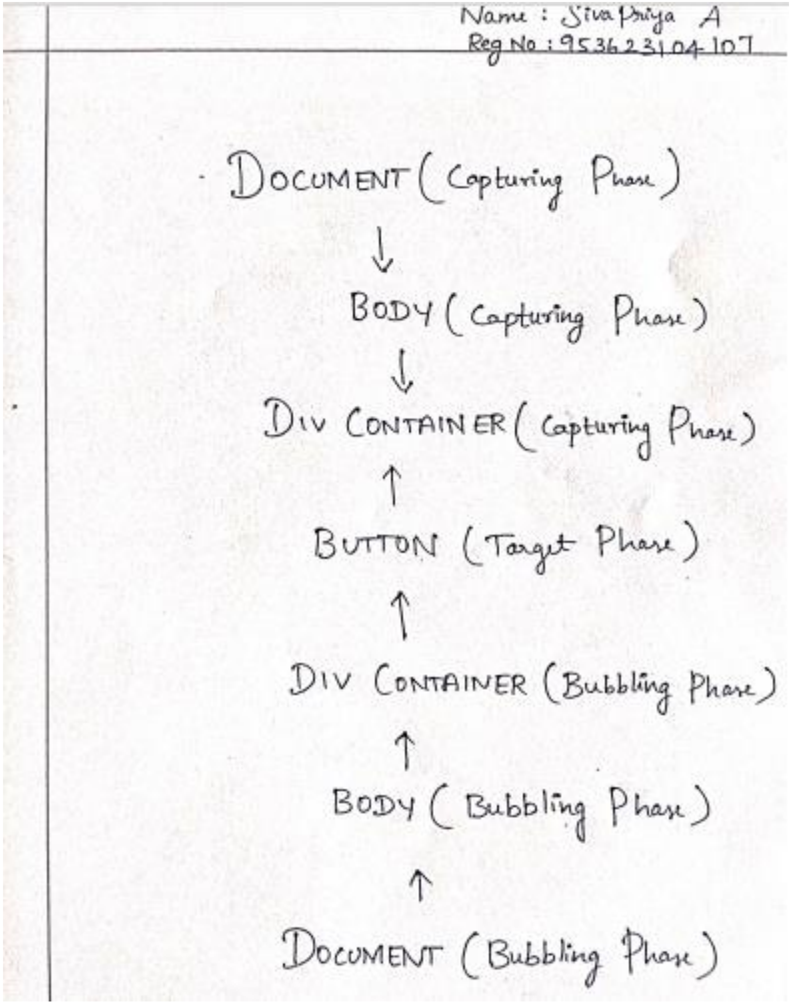

Fig:2. Activity by Siva Priya.A

Reflective Critique

- Students reported that the Think–Pair–Share structure helped them first build individual understanding, then refine it through peer discussion.
- The combination of reflection, collaboration, and demonstration deepened their conceptual clarity on how JavaScript handles events internally.
- Students understand how the activity linked theory to hands-on understanding, especially for event propagation (bubbling and capturing).
- The peer discussion phase reduced hesitation and encouraged active knowledge sharing among students.

Benefits of the Practice

- The activity promoted critical thinking and peer learning, allowing students to analyze and reason through event handling instead of memorizing it.
- The activity improved conceptual clarity regarding the relationship between the Event Source, Event Object, Listener, and Browser.
- The physical demonstration made abstract event flow processes more concrete. Students developed communication and collaborative skills, essential for group debugging and team-based web development projects.
- The Think–Pair–Share format catered to diverse learners those who learn best by thinking, discussing, or observing.

Challenges Faced in Implementation

- Some students initially struggled to articulate their individual understanding during the “Think” phase.
- Time management was a challenge, as pair discussions sometimes extended beyond the allotted duration.
- A few pairs were hesitant to share with the larger group; additional encouragement or gamification could increase participation.
- To improve inclusivity, future sessions could incorporate quick written reflections or digital polls during the “Think” phase to engage all students simultaneously.

References

- https://www.w3schools.com/js/js_events.asp
- https://developer.mozilla.org/en-US/docs/Learn/JavaScript/Building_blocks/Events
- https://www.geeksforgeeks.org/javascript-events/

Signature of Faculty Member **HOD**